

Research Papers

Numerical simulation of a medium-sized refrigerated truck box with low-temperature phase change cold storage material

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ABSTRACT

Cold chain transportation occupies a central position within the domain of domestic logistics, yet its significant energy expenditure in maintaining low temperatures within refrigerated containers necessitates urgent refinement. This study establishes a numerical model of a medium-sized refrigerated truck box using Fluent software, investigating the impact of various types, quantities, and infusion shapes of phase change materials (PCMs) on cold chain transport when integrated into the container. The results illustrate that the incorporation of PCM enhances the cooling efficacy of the system, resulting in a substantial 4 °C reduction in temperature fluctuations within the reefer truck. Particularly notable is the exceptional performance of RT3HC, which exhibits the most pronounced enhancement in air temperature stability. Furthermore, increasing the quantity of PCMs, while ensuring adequate air circulation between them, amplifies heat transfer efficiency and the rate of solidification/melting. Notably, the heat transfer efficiency of a configuration featuring 9 PCMs surpasses that of a 3 PCM arrangement by a remarkable 57.13 %. The implementation of a flow-guiding PCM configuration not only fosters temperature uniformity within the container, addressing slow cooling concerns in specific non-windward regions, but also enhances solidification efficiency by 3.8 % compared to the elongated strip-shaped PCM configuration. These findings furnish valuable insights for optimizing the cold chain transportation system, thereby enhancing its efficiency and sustainability.

1. Introduction

Amidst escalating global energy demands, low-temperature cold chain transportation has emerged as a pivotal trajectory within the domestic logistics industry. However, this advancement is counterbalanced by the challenge of maintaining low temperatures while striving for decarbonization. Notably, energy consumption associated with cold chain transportation contributes, both directly and indirectly, to approximately 2.5 % of global greenhouse gas emissions [1]. A substantial portion of this energy is allocated to maintaining low temperatures inside refrigerated containers, with refrigeration alone accounting for 60–70 % of the electrical energy consumption. Consequently, there is significant untapped potential for improving energy consumption and mitigating greenhouse gas emissions within cold chain transportation.

Research on refrigerated trucks has been ongoing. Song et al. [2] subsequently developed a simulation model of the refrigeration system

employing TRNSYS and MATLAB. The optimization of the energy consumption of the chiller's main operational parameters was achieved through genetic algorithms. Their findings revealed a noteworthy average reduction of 0.5 kW in the total energy consumption of the refrigeration system, coupled with an average increase of 0.15 in the Coefficient of Performance (COP). Wu et al. [3] introduced a novel carbon footprint model to assess refrigerated trucks, encompassing the life cycle impacts of the refrigeration system, the insulated truck body, and the truck chassis. Their study highlighted the substantial carbon footprint of refrigerated trucks during the usage phase, constituting approximately 96 % of the total greenhouse gas (GHG) emissions attributed to refrigerated trucks. Taher et al. [4] conducted numerical investigations into the aerodynamic and thermal performance of refrigerated trucks under various scenarios utilizing URANS. Their research demonstrated the efficacy of air curtains in optimizing energy consumption and cooling duration during the opening of truck doors across diverse loading scenarios. Utilization of air curtains resulted in potential energy savings of up to 25.3 % in cooling energy consumption.

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Nomenclature

List of symbols

A	Surface area (m^2)
c	Specific heat ($\text{J/kg}\cdot\text{K}$)
h	Enthalpy (kJ/kg)
h^*	Saturation enthalpy
k	Thermal conductivity ($\text{W/m}^2\cdot\text{K}$)
n	Amount of PCM
$\dot{q}$	Heat flux (W/m^2)
t	Time (s)
T	Temperature (K)
V	Volume (m^3)
v	Local velocity (m/s)

Subscripts

l	Liquid
m	Phase change
s	Solid

Greek letters

ρ	Density (kg/m^3)
β_{hs}	Converted liquid phase rate

Acronyms

COP	Coefficient of performance
GHG	Greenhouse gas
LTES	Latent thermal energy storage
PCMs	Phase change materials

Exposure of refrigerated truck boxes to sunlight during warm seasons can lead to a temperature increase. Consequently, scholars both domestically and internationally have focused on passive low-temperature distribution methods to address this issue by integrating phase-change cold-storage materials into the truck boxes. Fioretti et al. [5] explored the integration of PCM in refrigerated boxes, culminating in stabilized internal temperatures and reduced energy consumption. Ahmed et al. [6] innovatively improved conventional methods of insulating refrigerated trucks by utilizing copper tubes to encapsulate paraffin wax, uniformly arranged within Polyurethane foam. This approach effectively mitigated both peak heat transfer and total heat transfer. Michele et al. [7] proposed a novel concept by incorporating polyurethane foam enveloped with PCMs to create the Latent Thermal Energy Storage (LTES) system, subsequently examining its thermal performance through numerical simulation. Researcher Zhao et al. [8] advanced PCMs with a phase change temperature range of 2–8 °C, employing a blend of tetradecane and lauryl alcohol as the base fluid and integrating expanded graphite to enhance thermal conductivity. Mousazade et al. [9] investigated the influence of truck speed on the overall heat transfer coefficient by monitoring temperature fluctuations inside the box of a 6-ton refrigerated truck equipped with three PCMs. Their findings indicated that truck speed positively impacted the overall heat transfer coefficient. Phase-change cooling storage technology offers a reliable energy solution by utilizing phase-change materials (PCMs) to store or release thermal energy at specific temperatures. The stability of PCM temperature ensures the relative stability of the surrounding air temperature, thereby facilitating precise temperature regulation. Consequently, this technology is widely applied in fields requiring precise temperature control, such as cold chain transportation [10,11], building energy efficiency [12], and component thermal management [13].

The primary sectors for cold chain logistics are food and pharmaceuticals. Cold chain transportation of pharmaceuticals requires precise temperature regulation; otherwise, the effectiveness of the drugs may be

compromised. Composite phase-change cold storage materials are a promising solution due to their ability to meet stringent temperature requirements. By combining organic and inorganic components, these materials can effectively meet the demands of cold chain transportation. Numerous researchers have made significant contributions to this field. Bao et al. [14] successfully reduced the supercooling degree of industrial-grade PCM ($\text{CaCl}_2\cdot 6\text{H}_2\text{O}$) while enhancing its thermal conductivity and overall performance using flake graphite as a nucleating agent and a superabsorbent polymer as a thickener. Xiao et al. [15] pioneered nano-inorganic phase change “alloy” materials, investigating their exceptional properties and potential applications in the realm of building energy efficiency. Tang et al. [16] introduced a novel pseudo-ionic liquid PCM comprising CTAB and urea, well-suited for hot water supply in solar water heating systems, and achieved ideal phase change behavior through optimization of the mass ratio. On a different note, Bumsoo Han et al. [17] conducted investigations on the latent heat during the phase change process of water-NaCl binary mixtures, revealing that the size of the eutectic melting region increases with increasing NaCl concentration and reaches a maximum at the eutectic concentration. Furthermore, researchers such as Porisini et al. [18] focused on evaluating the corrosion resistance of various metal alloys to the hydrated salts of inorganic-based PCMs, with stainless steel demonstrating superior performance. Additionally, Diani et al. [19] explored the possibility of utilizing paraffin waxes with diverse melting temperatures embedded in aluminum foams with different pore sizes, significantly influencing the phase change process. In the quest for optimal materials for PCM containers, Eduard Oró et al. [20] investigated the corrosion effects of different metallic and polymeric materials in contact with nine phase change temperatures within the range of –22 °C to –16 °C, recommending the use of 316 stainless steel as an ideal material for PCM containers. In their study, Atinafu et al. [21] developed composites by incorporating multi-walled carbon nanotubes (CNTs) and biochar into bamboo, with n-dodecane serving as the energy storage material. Strategically placing the PCM on the surface of the evaporator stores some of the cold and solidifies it during operation of the refrigeration system. Subsequently, upon deactivation of the refrigeration system, the stored cold volume is released and liquefied to maintain a constant chamber temperature. Sonnenrein et al. [22] reported that refrigeration equipment integrated with PCMs significantly reduces power consumption, minimizes temperature fluctuations, and prolongs cooling cycles. Conversely, Alzuwaid et al. [23] investigated a cooler accumulator utilizing hydrogel PCM, positioned behind the evaporator as an auxiliary component to sustain temperature stability and achieve substantial energy savings. Copertaro et al. [24] placed the RT35HC PCM phase-change layer between the outer skin and the standard thermal insulation. Their study revealed that incorporating the phase change layer could reduce the peak heat load inside the refrigerated vehicle by 20.01 % to 20.87 % and decrease daily energy consumption by 4.55 % to 4.74 %, effectively lowering energy consumption and greenhouse gas emissions. Benoit Michel et al. [25] investigated the design methodology for a multilayer insulated wall used in refrigerated vehicles, which included a composite layer of polyurethane and PCM. In summary, while PCMs are primarily integrated into the shape of the shell of refrigerated trucks, there has been less research conducted on the insulation and energy-saving benefits of utilizing PCMs inside the box rather than solely in the shell.

To mitigate temperature fluctuations during loading and unloading operations, this study proposes a novel method of integrating a phase-change material (PCM) on top of a medium-sized refrigerated truck box. By absorbing and releasing heat during door openings, the PCM helps maintain the desired temperature inside the truck. Advanced numerical simulations, validated against experimental data, were employed to model the temperature distribution and cooling performance inside the truck, supporting further optimization studies. The study provides insights into optimizing PCM performance, identifying optimal phase transition temperatures and material configurations to

enhance energy efficiency. A detailed analysis of temperature distribution highlights the efficacy of PCM in reducing fluctuations, providing important insights into the design and optimization of refrigerated transport systems.

2. Numerical method

2.1. Physical model

Medium-duty reefer trucks serve as ground transportation refrigerated vehicles, where the box is constructed using insulated sandwich panels with steel construction to ensure robustness and strength. The insulation thickness typically ranges from 50 to 150 mm, specifically designed for short-haul refrigerated transportation, distinguishing them from long-haul trucks or tractor-trailers. In this study, the insulated sandwich panels used for the medium-sized reefer trucks have a thickness of 51.7 mm, with polyurethane foam insulation serving as the core element. These panels consist of galvanized outer layers and inner panels made of food-grade stainless steel. To maintain a visually appealing appearance, the outer layer of the refrigerated truck is coated with white automotive paint ES-170. Table 1 illustrates the composition of the wall surfaces of the refrigerated truck. The dimensions of the refrigerated truck box in this research are 3 m in length, 1.6 m in width, and 1.7 m in height, while the refrigeration system's dimensions are 1 m in length, 0.7 m in width, and 0.2 m in height. The refrigeration equipment used in the study is a THERMAL MASTER T-1400, with a cooling capacity of 4689 W at an ambient temperature of 30 °C and 2400 RPM. It is essential to note that the system under investigation solely experiences cooling; thus, the refrigeration equipment cannot reach the phase transition temperature, and no phase transition process occurs within the system.

Fig. 1 illustrates a simplified model of a medium-sized refrigerated truck cabinet [26]. The diagram includes four distinct temperature monitoring points: point 1 at the air inlet, point 2 at the air outlet, point 3 on the floor of the cabinet, and point 4 at the cabinet door.

2.2. Model-related assumptions

This paper presents a study on a medium-sized refrigerated truck box using several simplifying assumptions. The complex airflow inside the refrigeration system is disregarded, and an open system is adopted, using the enthalpy difference between the air inlet and outlet to approximate a closed system. The air inside the box is treated as a Newtonian incompressible fluid, while cold loss from the refrigeration equipment walls and ventilation slots on the floor are ignored. Heat transfer is assumed to be solely conductive and convective, neglecting radiative heat transfer. Air infiltration in gaps and associated cold loss are disregarded, and the impact of external airflow on the box's wall surface during travel is considered uniform. These assumptions enable simplified modeling and calculations, providing reasonably accurate predictions for the refrigerated truck system's behavior.

2.3. Phase change heat transfer mathematical model

Shamsundar et al. [27] introduced an enthalpy model as a solution to

Table 1
Wall composition of container bodies.

Categories of walls	Constituent elements	Thicknesses (mm)
Floorboards	Galvanized steel sheet	1
	Polyurethane foam board	50
	AISI 304 stainless steel plate	0.7
Side walls	Auto paint ES-170	0.1
	Galvanized steel sheet	1
	Polyurethane foam board	50
	AISI 304 stainless steel plate	0.7

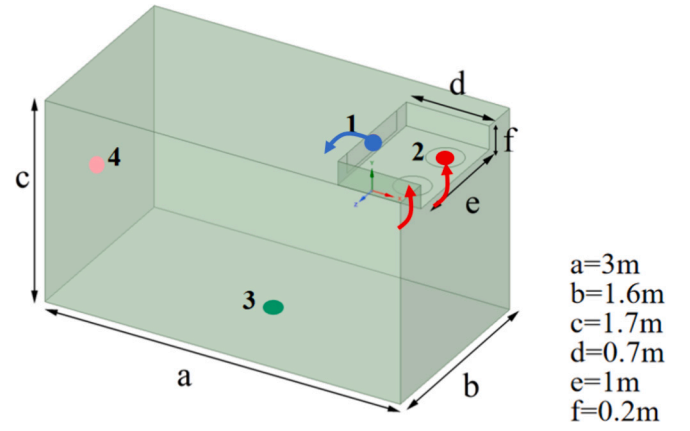


Fig. 1. Schematic shape of the box model of the medium-sized refrigerated truck.

multidimensional phase transition problems, where the mathematical expression of the model aligns with conventional conservation equations. This enthalpy approach model encompasses two dependent variables: enthalpy and temperature. The model formulates energy equations based on these two variables, covering solid-phase, liquid-phase, and two-phase regions. By employing numerical solutions for enthalpy-thermodynamic distributions, the model calculates the positions of the two-phase interfaces. The enthalpy approach model is suitable for addressing multidimensional phase change problems, as it avoids the necessity for distinct treatment of the solid and liquid phases. Given its compatibility with the present study, the enthalpy method is adopted to solve the research problem. The mathematical formulation for the enthalpy method model is as follows:

Given the small size and thickness of the phase change material utilized in this study, the convective heat transfer effect can be disregarded. Thus, the energy balance equation is established for any given control volume with surface area A and volume V , without accounting for convective heat transfer or an internal heat source:

$$\frac{d}{dt} \int_V \rho h dV + \int_s \rho h v \cdot dA = \int_s k \nabla T \cdot dA + \int_V \dot{q} dV \quad (1)$$

When the solid-phase specific heat (c_s) and liquid-phase specific heat (c_l) are constants, the expressions for enthalpy (h) and temperature (T) can be represented as follows:

$$T - T_m = \begin{cases} (h - h_s^*)/c_s & h < h_s^* \\ 0 & h_s^* \leq h \leq h_l^* \\ (h - h_l^*)/c_l & h_l^* < h \end{cases} \quad (2)$$

Without regard to convection and without a heat source:

$$\rho \frac{\partial h}{\partial t} = k \nabla^2 T \quad r \in V \quad (3)$$

$$\rho = \begin{cases} \rho_s & h < h_s^* \\ \rho_l & h > h_l^* \end{cases} \quad k = \begin{cases} k_s & h < h_s^* \\ k_l & h > h_l^* \end{cases} \quad (4)$$

where h_s^* and h_l^* are the enthalpies of saturation of the solid and liquid phases, respectively; and T_m is the phase transition temperature.

2.4. Numerical simulation solver setup

The objective of this study is to perform a numerical simulation of a medium-sized refrigerated truck box by employing Mesh software for meshing and setting up Fluent to meet operational requirements. Setting up the Fluent solver is a crucial aspect with several steps, each impacting the simulation results. First, in the mesh checking stage, ensuring good

quality and appropriate size of the mesh is essential. The transient model is enabled, and the effects of gravity are considered. Second, appropriate models are selected to depict energy transfer and fluid turbulent motion characteristics during the simulation. In this study, the Energy model and the SST $k-\omega$ model were selected to precisely evaluate regions close to and far from the wall.

The study involves air, aluminum, and polyurethane foam panels. Computational domain settings include defining the fluid domain as air and the solid domain as the insulation layer. Boundary conditions are set to account for convective heat transfer between the external wall surface of the refrigerated truck box and the external air, excluding external radiative heat transfer. For the non-slip wall surface, a convective boundary condition is applied, with the convection coefficient calculated for each wall surface as $3.41 \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$ [26]. The wall material is redefined as polyurethane foam board. Since the cold dissipation of the wall surface of the refrigeration equipment is not considered, a heat flux boundary condition is set for this wall surface, with the heat flux set to $0 \text{ W}/\text{m}^2$, making it an adiabatic wall surface. The air outlet of the refrigeration equipment is defined as a velocity inlet, with the ventilation condition during the closure of the box door specified as an average inlet velocity of 3 m/s . The return air outlet is set as a free-flow outlet. Considering that the temperature of the vehicle is not constant, the refrigeration equipment outlet temperature must be input into the software as an input function (temperature versus time), necessitating dynamic simulation using Fluent, where a user-defined function (UDF) becomes essential. In this study, the time-dependent variation values of the experimental data were fitted by Origin to obtain a function of the air outlet temperature over time, which was then imported into Fluent using the UDF with the following functional expression [26]:

$$T(t) = 11.95 \cdot \exp(-0.001058 \cdot t) + 11.58 \cdot \exp(-0.01036 \cdot t) \quad (5)$$

For the solution methodology, the SIMPLE algorithm is employed, assuming independence between the velocity and pressure fields while correcting the velocity field. Spatial discretization uses different difference schemes to ensure calculation accuracy. Data monitoring setups are aligned with experimental monitoring points, collecting data using temperature sensors to validate the model's efficacy. Finally, initializing

the temperature to ambient temperature, and setting the number of calculation steps and time steps for iterative calculation.

3. Calculation verification

3.1. Mesh segmentation and irrelevance verification

An unstructured mesh is used for the division of the mesh inside the box. To streamline the calculation process and reduce unnecessary meshes, the thickness of the refrigerated truck box wall was not specified in SpaceClaim. Consequently, there was no need to mesh the polyurethane foam insulation wall in Mesh. Instead, the thickness of the polyurethane foam insulation wall was defined in Fluent for further calculations. The result of the meshing process is illustrated in Fig. 2.

Optimizing the grid resolution through grid-independence validation not only ensures computational accuracy but also enhances computational efficiency, thereby reducing computation time. In this study, grid-independence verification was conducted using different grid sizes, including models with 24,373, 106,258, 150,853, 227,248, and 271,255 grids. The solver settings used were consistent with those presented in Section 6. The objective of the validation was to analyze the time-averaged temperature of the air inside the refrigerated truck box. This verification process allows for the identification of the grid size that provides adequate accuracy while maximizing computation efficiency.

Based on the simulation data presented in Fig. 3, it is evident that the average temperature of the air inside the refrigerated truck box shows a decreasing trend with an increase in the number of grids, indicating improved accuracy. However, beyond a certain grid count, specifically at 227,248 grids, the average temperature of the air shows only minor changes with further increases in grid count. Comparing this grid count with 271,255 grids, the maximum error in the time-averaged temperature is only 0.02°C . Therefore, it can be concluded that the data obtained from simulations with 227,248 grids are sufficiently accurate. There is no substantial gain in accuracy by increasing the grid count to 271,255. Hence, for subsequent calculations, the grid can be divided in the same manner as in the simulations with 227,248 grids. This choice ensures a balance between accuracy and computational efficiency,

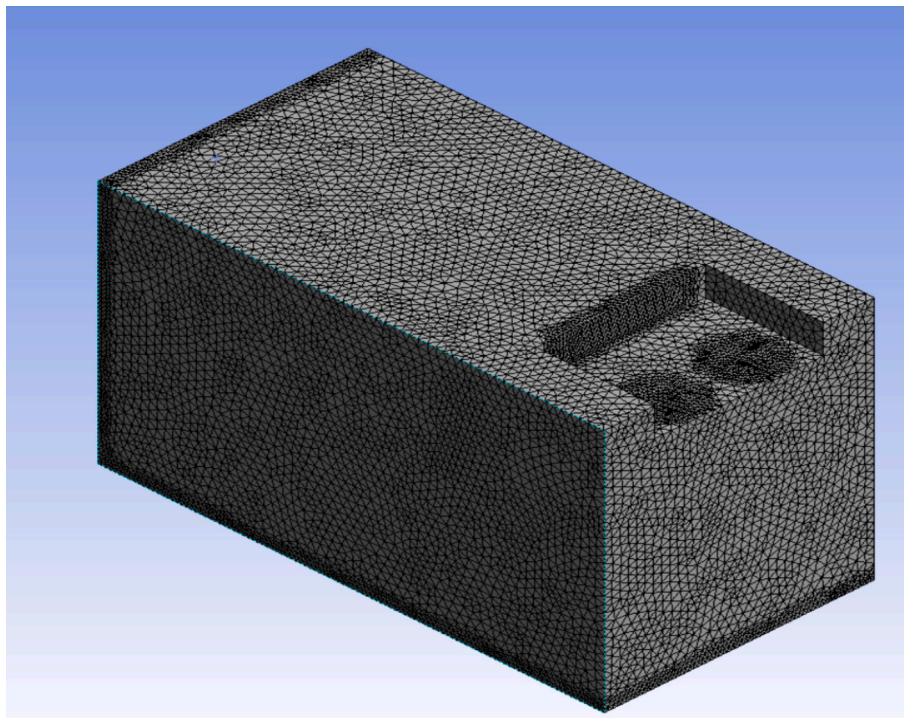


Fig. 2. Medium-sized refrigerated truck box mesh delineation.

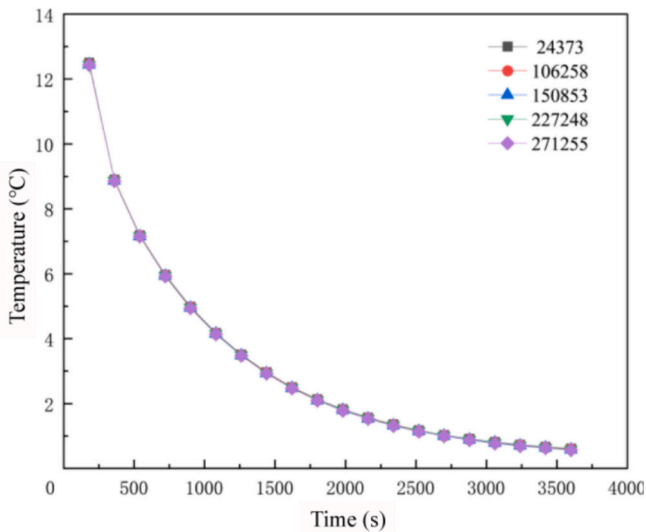


Fig. 3. Relationship between average temperature and different grids.

facilitating reliable results without unnecessary computational overhead.

3.2. Comparison with experimental data

To verify the accuracy of the model, the boundary conditions were set based on the methodology established by Jara et al [26]. The refrigeration equipment used R134A refrigerant and required 2965 W to maintain an internal cabinet temperature of 0 °C. The air temperature from the refrigeration equipment varied over time and was expelled at an average speed of 3 m/s, with the ambient environment having an average wind speed of 2.24 m/s. Ambient humidity was set at 60 %, and solar radiation intensity was 394.3 W/m². The verification of the model focused on time-averaged temperatures at four measurement points inside the vehicle's body.

Fig. 4 compares simulation and experimental data at four monitoring points: air inlet, air outlet, floor, and cabinet door. The experimental cooling process involved heat conduction, convection, and radiation. However, for simplicity in numerical simulation, radiative heat transfer was neglected, leading to deviations between simulated and experimental data. The experimental minimum temperature inside the box was higher than simulated due to this omission. Comparison of experimental and simulated air temperatures showed average deviations of 0.14 °C, 2.20 °C, 1.86 °C, and 0.92 °C for the four points, respectively. Despite these discrepancies, simulated temperatures mirrored experimental trends, with a gradual decline in rate. Thus, the numerical model accurately predicts temperatures inside the refrigerated truck. In conclusion, this study's numerical model effectively forecasts experimental outcomes and is suitable for further simulations. However, future work should consider including radiative heat transfer for enhanced accuracy.

4. Result

The model's validity is confirmed through comparison of experimental and simulation data, demonstrating its accurate predictive capabilities. Subsequently, to mitigate temperature fluctuations during loading and unloading in transportation trucks, a strip-shaped phase change cold storage material is added to the top of the medium-sized reefer box based on the initial model. This modification examines the impact of PCM on the temperature distribution inside the box and the system's cooling capacity recovery during open- and closed-door states. Furthermore, this chapter investigates the operational performance of

the system with various PCMs and determines the optimal phase change temperature for the transportation cycle. To evaluate the energy-saving benefits more effectively, the concept of "system's recovery cooling capacity" is introduced. This concept evaluates the energy required for the system to return to refrigerated temperatures at the end of a transportation cycle. A strong recovery capacity indicates robust energy-saving benefits, while a weak recovery capacity suggests limited energy-saving potential or none at all. This concept is applicable throughout subsequent chapters, providing a comprehensive evaluation of the system's energy efficiency.

4.1. Changes of phase change cold storage materials on the internal temperature field of refrigerated truck boxes

The next focus is on examining the impact of phase change cold storage materials on the temperature distribution and the system's cooling recovery within a refrigerated box under varying operational conditions. To achieve this, a strip-shaped PCM is added to the top of the medium-sized refrigerated truck box, maintaining its original dimensions without PCM. During the closed-door period, the refrigeration system runs to replenish the PCM's cooling capacity, ensuring readiness for the subsequent transportation cycle. Conversely, during the open-door period for unloading, the refrigeration system is turned off, allowing external heat to enter the box. The PCM then absorbs this heat, effectively maintaining the desired temperature inside the box. The study focuses on a single transportation cycle, with the vehicle remaining closed for 120 min and an additional 30 min with the door open during unloading (as depicted in Fig. 5).

4.1.1. Physical modeling of a medium-sized refrigerated truck box assembled with phase change cold storage materials

The physical model of the system maintains the same dimensions and overall structure as the refrigerated box without PCM. The only alteration, depicted in Fig. 6, involves adding a series of parallel hexagonal bars loaded with PCMs fixed to the ceiling of the refrigerated truck box. These PCM-loaded bars are primarily designed to minimize temperature fluctuations during loading and unloading processes. The geometric dimensions of these long PCM bars are as follows: they are 2 m long, 0.03 m wide, and 0.1 m high. To ensure efficient thermal conductivity and enhance the PCM's charging and cooling processes, the thin shell of these bars is made from lightweight aluminum, with a thickness of 2 mm. The use of aluminum also facilitates easy processing. Additionally, intentional spacing between the bars allows for adequate air circulation within the system. This design feature aims to optimize contact between the cold air and the PCM, thereby ensuring effective temperature regulation and distribution inside the refrigerated truck box.

4.1.2. Thermophysical parameters of phase change cold storage materials

In selecting phase change cold storage materials for this model, several critical factors are considered. Given that refrigerated trucks undergo multiple loading and unloading cycles during cold chain transportation, it is essential for the PCMs inside the box to possess excellent chemical properties, stability, and reversibility. Organic PCMs are known for their superior stability compared to inorganic and composite PCMs [28], making them more suitable for use in refrigerated trucks. Additionally, the phase change temperature of the PCM must be carefully chosen to ensure efficient operation. Considering the temperature fluctuation range inside the refrigerated truck box, which spans from 0 to 22.5 °C, selecting a phase change temperature slightly above 0 °C allows for solidification during the closed-door period. Similarly, the phase change temperature should not be excessively high to facilitate melting during door opening. Based on these considerations, this study employs paraffin wax as the phase change cold storage material. Paraffin wax offers various types, and for enhancing the phase change process during transportation cycles, this research specifically uses the thermal properties of paraffin RT8HC for numerical simulations.

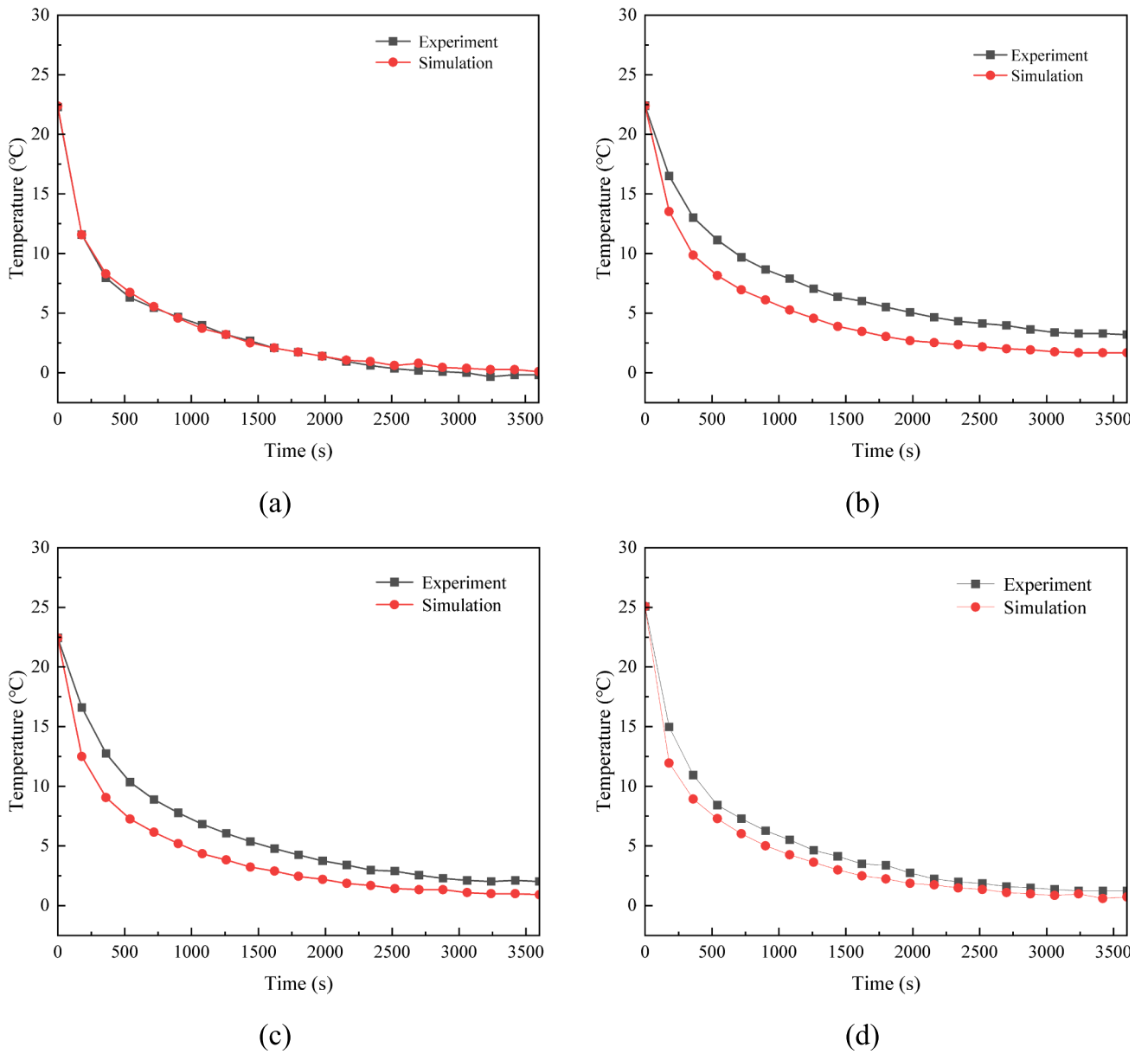


Fig. 4. Comparison between experimental and simulated time-by-time temperatures at four monitoring points inside a refrigerated truck box. (a) Monitoring point 1; (b) monitoring point 2; (c) monitoring point 3; (d) monitoring point 4.

(Table 2). This selection is anticipated to optimize temperature regulation and improve energy efficiency in refrigerated truck boxes during cold chain transportation.

4.1.3. PCM meshing

Meshing was conducted using ANSYS 2020R1 software. The meshing approach for the medium-sized refrigerated truck case mirrors the method described in Section 3.1, employing an unstructured mesh. However, due to the parallel hexahedral regular shape of the aluminum shell loaded with PCM, a structured mesh was determined to be more appropriate. Visual results of the PCM meshing are depicted in Fig. 7.

4.1.4. Model-related assumptions

Section 2.2 established assumptions for the medium-sized refrigerated truck box without the PCM assembly. In this study, specific assumptions are derived for the phase change cold storage material integrated into the medium-sized refrigerated truck box, building upon the original assumptions. The PCM is assumed to exhibit isotropic and

homogeneous characteristics, with stable and reversible phase change behavior and no subcooling effect. Thermal resistance at the PCM-aluminum shell interface is neglected, and the thickness of the aluminum shell is omitted for simplicity. The PCM's density remains constant in the buoyancy term of the momentum equation, employing Boussinesq's assumption.

4.1.5. Numerical model solver setup

The solver settings are modified based on the original refrigerated truck cabinet without PCM, focusing on the closed-door state. The following adjustments and configurations are made:

1. Grid Check: The same grid checking process as in Section 3.1 is performed.
2. Model Selection: The Energy model, SST k- ω model, and Solidification/Melting model are enabled in the Models category to cover the energy transfer, fluid turbulent motion, and solidification/melting processes of PCMs.

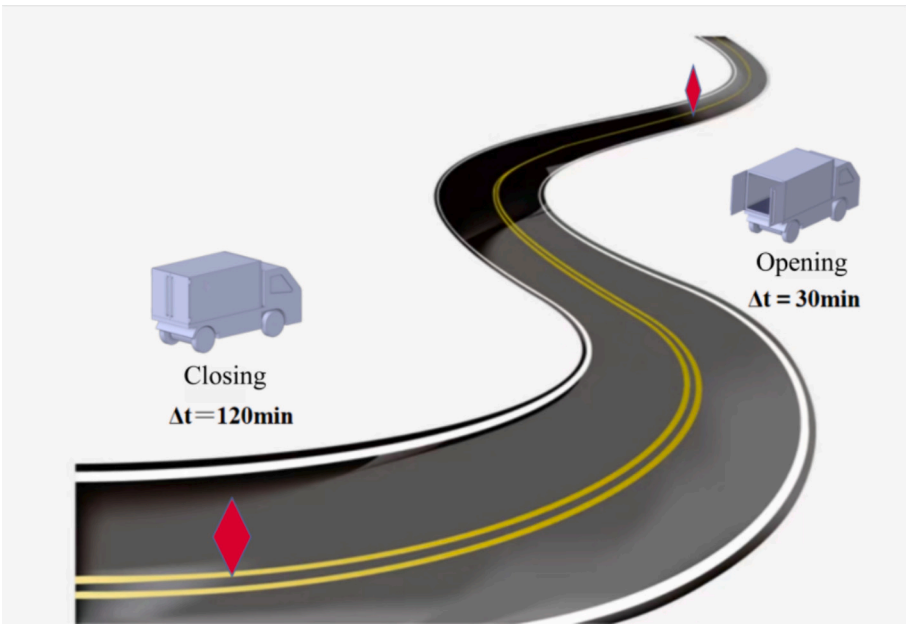


Fig. 5. Medium reefer transportation cycle.

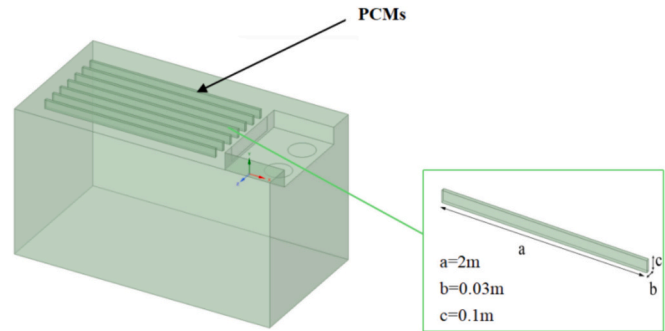


Fig. 6. Schematic shape of the box model of a medium-sized refrigerated truck equipped with a PCM.

Table 2
Thermophysical parameters of the PCM RT8HC [29].

Paraffin	RT8HC
Freezing point (°C)	8
Latent heat of phase transition (kJ/kg)	190
Densities (kg/m ³)	
Solid state	880
Liquid	770
Specific heat capacity (kJ/(kg·K))	
Solid state	2
Liquid	2
Thermal conductivity (W/(m·K))	
Solid state	0.2
Liquid	0.2

3. Material Setup and Selection: The PCM material, paraffin RT8HC, is defined with its properties, such as density, specific heat, thermal conductivity, and viscosity, as shown in Table 3. The density is assumed constant using Boussinesq with a value of 770.
4. Calculation Domain Settings: The fluid domain is designated for air, the solid domain for insulation, and the phase change domain for PCM RT8HC, with the corresponding material selections.
5. Boundary Conditions: The same boundary conditions as in Section 2.4 are set.
6. Solution Method: The Coupled algorithm is used for pressure-velocity coupling, solving the momentum equation for velocity, the mass conservation equation for pressure, and the energy

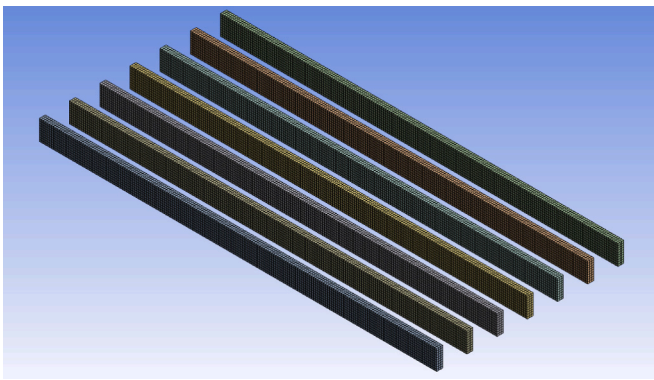


Fig. 7. Demonstration of PCM lattices.

Table 3
Thermophysical parameters of PCMs RT3HC, RT5, RT8HC, RT10HC [29].

Paraffin	RT3HC	RT5	RT8HC	RT10HC
Freezing point (°C)	3	5	8	10
Latent heat of phase transition (kJ/kg)	190	180	190	200
Densities (kg/m ³)				
Solid state	880	880	880	880
Liquid	770	770	770	770
Specific heat capacity (kJ/(kg·K))				
Solid state	2	2	2	2
Liquid	2	2	2	2
Thermal conductivity (W/(m·K))				
Solid state	0.2	0.2	0.2	0.2
Liquid	0.2	0.2	0.2	0.2

- conservation equation for temperature. Second-order windward difference format is used for spatial discretization, and default values are used for relaxation factors.
7. Data Monitoring: Monitoring focuses on the time-to-time changes of air temperature, PCM temperature, and PCM liquid phase rate in the chamber. Volume average of air and PCM temperature in the box and mass average of PCM liquid phase rate are reported with a monitoring interval of 180 s.

8. Initialization and Iteration Settings: The entire refrigerated truck box is initialized with a temperature of 295.65 K, and the PCM is locally initialized with a temperature of 285.15 K. The time step is set to 1 s, with 7200 computational steps during the door closing and 1800 steps during the door opening, resulting in a total simulation time of 9000 s.

4.1.6. Analysis of numerical simulation results

Fig. 8 illustrates a cloud map depicting the temperature distribution

dynamics during operational phases of a refrigerated truck box integrated with Phase Change Material (PCM). The temperature distribution along the median surface is monitored at discrete time intervals: 0 s, 3600 s, 7200 s, 7230 s, 7380 s, and 9000 s. Initially, the PCM registers a temperature of 285.15 K, while the internal air temperature measures 295.65 K. After one hour, the internal air temperature stabilizes, resulting in a uniform temperature distribution. Concurrently, the PCM temperature decreases to approximately 280 K, indicating the onset of solidification. By the two-hour mark, upon reaching its destination, the

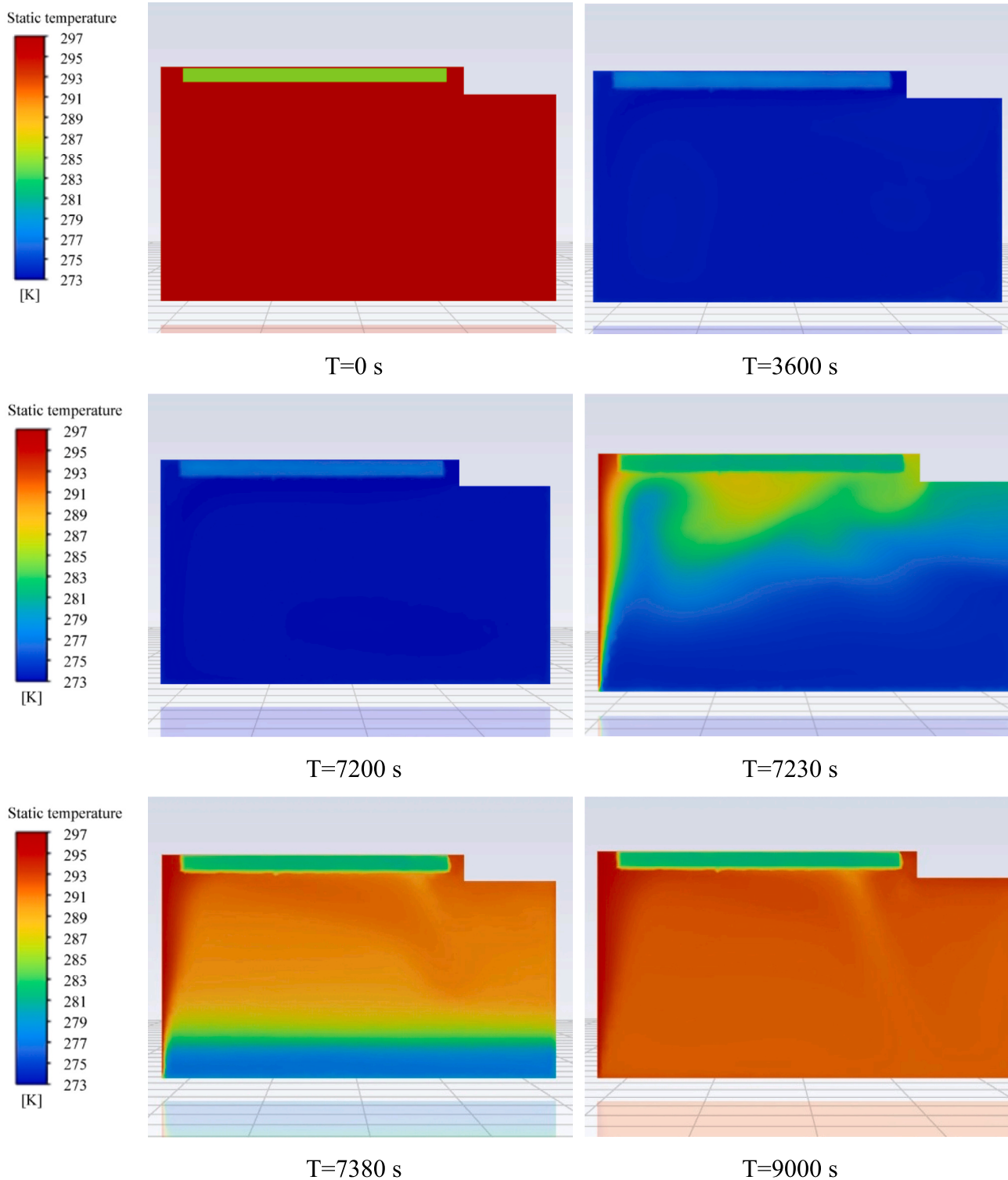


Fig. 8. Cloud view of temperature distribution during opening/closing of refrigerated truck box fitted with a PCM.

refrigerated truck initiates the door-opening sequence, reflected in the temperature profile observed at 3600 s. The ensuing natural convection, driven by the temperature differential between the internal and external environments, causes a rapid rise in internal temperature due to the influx of higher-temperature ambient air. This induces localized melting of the solid PCM as it absorbs heat. Additionally, density differences create a noticeable “upper hot and lower cold” temperature gradient within the box, as evidenced in the cloud diagrams at 7230 s and 7380 s. Ultimately, upon completion of unloading, the internal temperature stabilizes around 292 K, significantly elevated from its initial state.

Fig. 9 illustrates temporal variation curves showing the dynamics of air temperature inside a refrigerated truck box with and without Phase Change Material (PCM) across the transportation cycle. During the closing phase, corresponding to the activation of the refrigeration system, the internal temperature gradually decreases. The cooling process initially exhibits maximum cooling rate, gradually tapering off until it stabilizes around 3600 s. In contrast, during the opening phase, corresponding to the deactivation of the refrigeration system, the internal temperature experiences a rapid increase. The rate of temperature rise peaks initially and then slows down until stability is reached around 8100 s. A comparative analysis between operational scenarios with and without PCM reveals reduced temperature fluctuations within the refrigerated truck. Specifically, the inclusion of PCM mitigates temperature variations by approximately 4 °C during opening and closing cycles, thereby enhancing the system's capability to maintain temperature stability during cold chain transportation.

Fig. 10 presents time-dependent curves illustrating the liquid phase change rates for each of seven Phase Change Materials (PCM1 to PCM7) and their combined rate. The curves exhibit a similar pattern: an initial steady phase, followed by a decline and subsequent increase. Notably, PCM2 undergoes phase change first, whereas PCM1 and PCM7 are the last to complete their phase transitions. PCM2 shows the lowest liquid-phase rate, while PCM7 demonstrates the highest. During the shutdown period, PCM2 exhibits 47.3 % greater efficiency in cooling storage capacity compared to PCM7. The effectiveness of PCMs in cooling storage correlates directly with the exposure of their sidewalls to cold air, suggesting that positioning PCMs opposite the air outlet optimizes their performance. When considering all PCMs together, the combined liquid phase rate reaches its minimum at 0.6967. Moreover, during the unloading phase, only 2.37 % of the PCM melted, effectively minimizing temperature fluctuations inside the box.

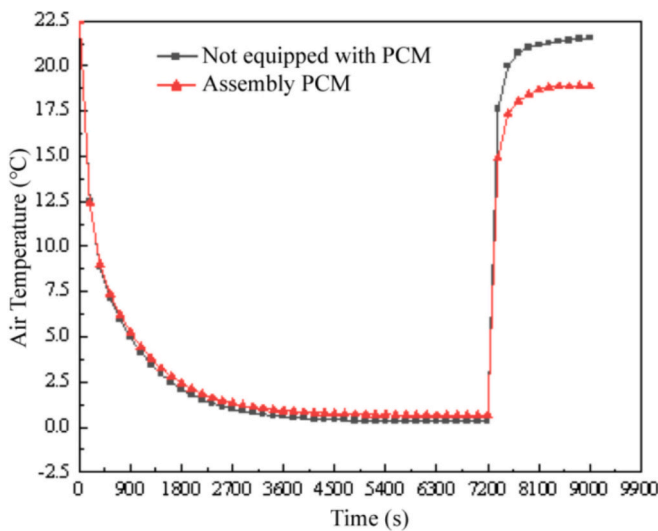


Fig. 9. Time-to-time variation curves of air temperature inside the refrigerated truck box with and without PCM assembly.

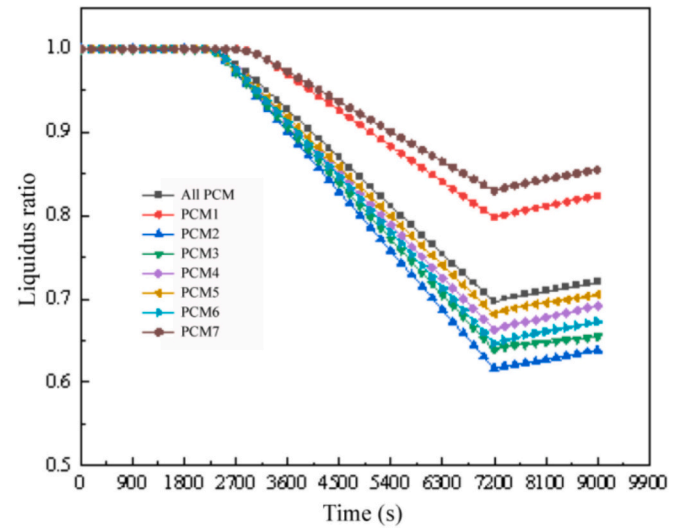


Fig. 10. Time-dependent variation curve of liquid phase rate of PCM RT8HC.

4.2. Changes of different phase change cold storage materials on the internal temperature field of a refrigerated truck box

In Section 4.1, the simulation results demonstrated that integrating PCMs inside the refrigerated truck box effectively mitigated temperature fluctuations during the opening and closing phases, while also enhancing the system's recovery cooling capability. To further investigate the influence of PCMs with varying phase change temperatures on system performance, this study conducted numerical simulations using PCMs with phase change temperatures RT3HC, RT5, RT8HC, and RT10HC. The configuration of PCMs—including their shape, quantity, and installation position—remained consistent across these simulations. The physical model, mesh division, model assumptions, and solver settings for these four conditions were established as described in Section 3.1 to maintain consistency and facilitate comparative analysis.

4.2.1. Thermophysical parameters of phase change cold storage materials

In Section 4.1.2, PCMs paraffin RT3HC, RT5, RT8HC, and RT10HC were selected for a comparative study based on the principles of PCM selection mentioned earlier. The thermophysical parameters of these four materials, each with different phase change temperatures, are presented in Table 3.

4.2.2. Analysis of numerical simulation results

Fig. 11 illustrates the time-dependent variation curves of the internal air temperature within refrigerated trucks equipped with different PCMs characterized by various phase change temperatures: RT3HC, RT5, RT8HC, and RT10HC. The curves detail the temperature profiles under four distinct operational conditions for each PCM. PCMs experience a sharp drop in temperature before the phase change because it has a low heat capacity in the single-phase state. The small range of temperature variation during the phase change is due to the fact that the material absorbs or releases a large amount of latent heat of phase change and the temperature remains essentially constant. At 7200 s, the air temperature with RT3HC reaches its minimum value at 0.43 °C, whereas with RT10HC, it peaks at 0.7429 °C, indicating a maximum temperature disparity of 0.3129 °C between these PCMs at this interval. During the opening period, a rise in internal air temperature is observed. By 9000 s, the air temperature with RT3HC remains the lowest at 18.3151 °C, while the temperature with RT10HC reaches the highest at 19.287 °C, resulting in a temperature difference of 0.9719 °C. These observations underscore that the RT3HC condition exhibits superior cooling recovery capability compared to RT5, RT8HC, and RT10HC under identical conditions. Over the entire transportation cycle, the RT3HC condition

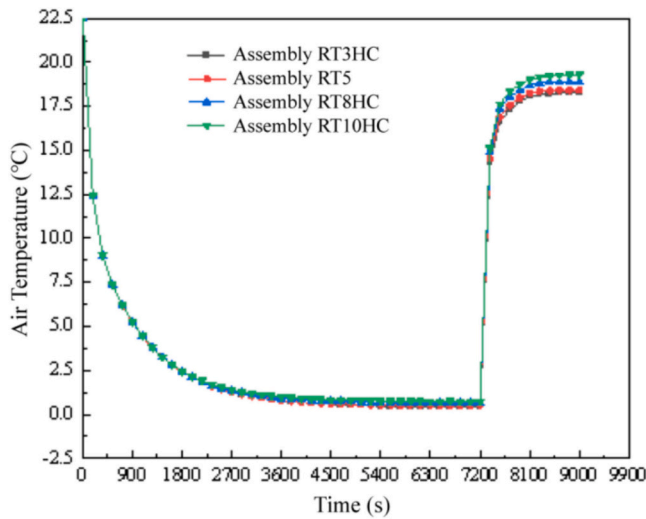


Fig. 11. Time-to-time variation curves of internal air temperature of refrigerated trucks fitted with PCMs of different phase change temperatures.

markedly reduces air temperature fluctuations compared to the other three conditions, thereby demonstrating the most effective performance.

Figs. 12 and 13 illustrate the temperature and liquid phase rate curves of the PCMs under four different working conditions, respectively. Analysis of the liquid phase rate change curve reveals that the melting rate of PCM during the door opening period is lower than the solidification rate during the door closing period. This disparity is attributed to the dominance of forced convection during the door closing period, leading to more intense cooling, whereas natural convection dominates during the door opening period. Throughout the single transportation cycle, none of the four PCMs fully solidified or melted. RT10HC, with the highest phase change temperature, solidified first at 1440 s, whereas RT3HC began its phase change only at 4500 s. During the shutdown period, RT3HC exhibited the highest phase change temperature. The heat transfer efficiency of RT10HC was 87.56 % higher than that of RT3HC during the door closing period due to the difference in phase change temperature. RT5 and RT8HC, with phase change temperatures between those of RT3HC and RT10HC, demonstrated higher heat transfer efficiencies than RT3HC but lower than RT10HC. During the door opening period, due to natural convection and the short

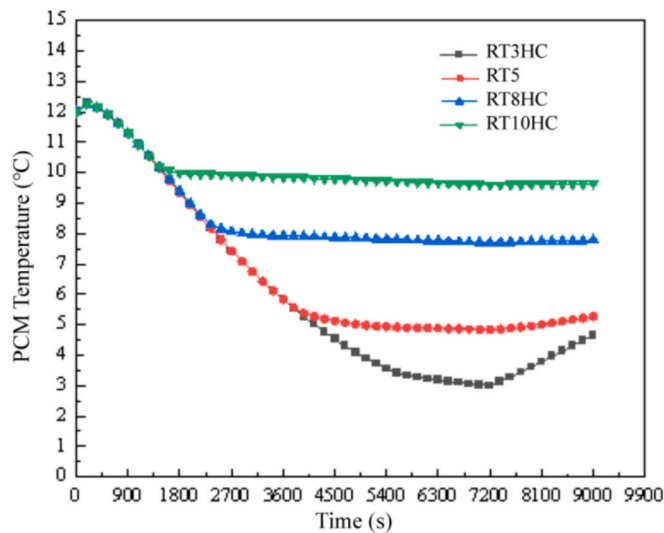


Fig. 12. Time-dependent variation curves of PCM temperature under four operating conditions.

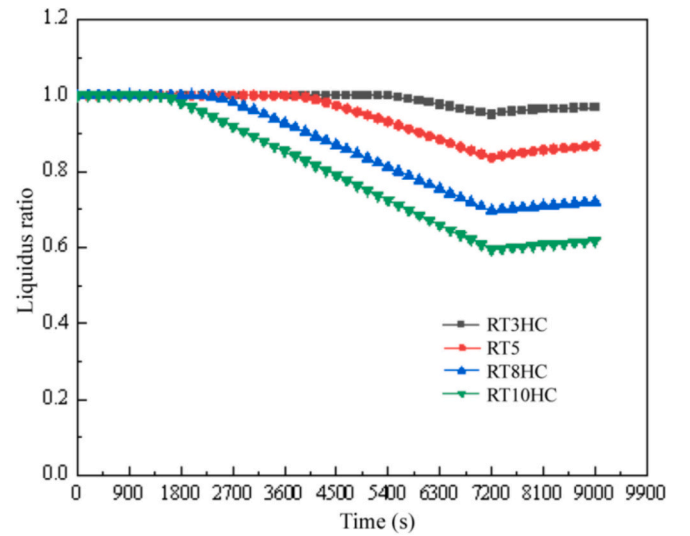


Fig. 13. Time-dependent variation curves of liquid phase rate of PCMs under four operating conditions.

unloading time, the heat transfer efficiency advantages of RT5, RT8HC, and RT10HC were not as pronounced. The liquid phase rate change curve of RT3HC indicated sufficient heat transfer efficiency to solidify during the 7200 s of the door closing period and to melt and absorb part of the heat during the door opening period. RT3HC maintained a temperature of 3–4 °C during the phase change, effectively lowering the surrounding air temperature. Consequently, RT3HC exhibited a higher recovery cooling capacity compared to the other three PCMs.

5. Optimization

The study further investigated the impact of the quantity and shape of long phase change materials (PCMs) on reefer box performance. Leveraging the RT8HC operating conditions of PCMs, we varied the number of strip-shaped PCMs and conducted simulation analysis following a rational layout. Furthermore, we optimized the shape of long strip PCMs by implementing a flow-guiding design. While maintaining identical types and quantities of PCMs, we compared and analyzed the heat exchange efficiency between the two shapes and their influence on the internal temperature field of the refrigerated truck box. This research aims to identify a more advantageous assembly scheme for PCMs.

5.1. Changes of the amount of phase change cold storage material on the internal temperature field of a refrigerated truck box

This section examines how the quantity of phase change cold storage materials affects temperature distribution and system recovery cooling capacity inside the refrigerated truck box under varying door conditions. The study utilizes a refrigerated truck box equipped with PCM RT8HC, detailed in Section 3.1. We modify the physical model by adjusting the number and arrangement of PCMs while keeping the size of the strip-shaped PCM and the refrigerated truck box constant. The mesh size, model assumptions, and solver settings remain unchanged from the original model to ensure a fair and accurate comparison.

5.1.1. Physical modeling of a medium-sized refrigerated truck box assembled with different amounts of PCMs

In this section, we investigate four distinct operating conditions using the physical model of the refrigerated truck box equipped with PCM. These conditions entail varying the quantity and layout of strip-shaped PCMs while maintaining the overall framework of the physical model. The four conditions are: 3 PCMs, 5 PCMs, 7 PCMs (as described in

detail in Section 3.1), and 9 PCMs. The physical model for the 7 PCMs condition has been detailed in Section 3.1 and is not reiterated here. Figures depicting the physical models for the 3, 5, and 9 PCMs conditions are shown in Fig. 14.

5.1.2. Analysis of numerical simulation results

Fig. 15 displays cloud diagrams showing top-view temperature distribution within the refrigerated truck box under different quantities of PCMs. Temperature distributions were recorded at top-view planes at 7200 s and 9000 s for four different conditions. Despite similar overall temperature distribution trends across these conditions, a notable observation is the increasing impact on air temperature within the material gaps with higher numbers and densities of PCMs. For example, in the setup with 9 PCMs, the reduced gap distance results in the air temperature around the PCMs closely matching the PCM temperature. This suggests that with adequate airflow within the gaps, an increased number of PCMs significantly affects the surrounding air temperature during the phase change process. Thus, denser PCM distribution improves thermal regulation within the refrigerated truck box by aligning air temperature more closely with PCM temperature during cooling and heating phases.

Fig. 16 illustrates the time-dependent variation curves of the internal air temperature in a refrigerated truck equipped with varying numbers of PCMs. The four curves represent configurations with 3, 5, 7, and 9 PCMs, respectively. Observations indicate that the temperature change trends for these four conditions are consistent, especially before 7200 s, where the curves nearly coincide, and the maximum temperature difference is a mere 0.30 °C, with the 9 PCM condition having the highest temperature and the 3 PCM condition the lowest. This can be attributed to the broader influence of the PCM temperature as the quantity increases. The phase change temperature of 8 °C was used for the target cooling temperature of the 0 °C refrigerated truck box. As the number of PCMs increases, the temperature influence range expands, leading to a slight rise in the average room temperature. At 7200 s, the air temperature for all conditions increases sharply, initially at the highest rate, before gradually leveling off. However, the temperatures at which the four conditions reach a steady state differ: the 9 PCM condition stabilizes at the lowest temperature of 18.51 °C, while the 3 PCM condition stabilizes at the highest temperature of 19.91 °C, with the 5 PCM and 7 PCM conditions falling in between. Considering both the closing and opening phases, the 9 PCM configuration shows the greatest improvement in reducing temperature fluctuation, lowering the temperature variation inside the refrigerated truck by 1.7 °C compared to the 3 PCM condition, demonstrating the best performance. This improvement can be attributed to two main factors: firstly, 9 PCMs can store or release colder within the same time frame, reducing heat infiltration during door opening. Secondly, with more PCMs and smaller gaps between them after re-layout, the airflow rate within the gaps increases, enhancing heat exchange between the air and the PCMs. This accelerates the cold storage process in the phase-change materials, significantly boosting the system's recovery cooling capacity.

Fig. 17 shows the time-dependent variation curves of the liquid phase rate for different quantities of PCMs. It can be observed that for

the four conditions with different quantities of PCMs assembled, the liquid phase rate of the 3 PCM condition is the lowest, indicating the highest utilization rate of PCM, while the 9 PCM condition has the highest liquid phase rate, indicating the lowest utilization rate of PCM. The 5 PCM and 7 PCM conditions fall in between these two extremes in one transportation cycle. Since the different working conditions involve different masses of PCMs, directly comparing the liquid phase rates may not provide a fair assessment of their heat transfer efficiencies. Therefore, to make a fair comparison, this study needs to convert the liquid phase rate to unify the PCM mass. The liquid phase rate conversion formula is based on the working condition of 9 PCMs as a reference standard, and the specific mathematical expressions are as follows:

$$\beta_{hs} = 1 - \frac{n}{9}(1 - \beta) \quad (6)$$

where β_{hs} is the converted liquid phase rate; n is the amount of PCM.

Fig. 18 depicts the time-dependent variation curves of the liquid phase rate after standardizing the PCM mass across four distinct working conditions. The trends in liquid phase rate change exhibit consistency across all cases. Beginning at 2160 s, all configurations commence solidification, with the 9 PCM configuration displaying the swiftest solidification rate and the 3 PCM configuration exhibiting the slowest. At 7200 s, following door closure and completion of transportation, the liquid phase rate reaches its nadir across all configurations. Notably, the heat exchange efficiency of the 9 PCM configuration surpasses that of the 3 PCM configuration by 57.13 % at this juncture. Subsequently, post-7200 s, as the door is opened for unloading, PCMs commence heat absorption and melting, resulting in an uptick in the liquid phase rate across all conditions. The melting rate of the 9 PCM configuration marginally outpaces that of the other three configurations. In summary, augmenting the number of PCMs while ensuring unimpeded air circulation between them enhances material heat transfer efficiency and bolsters solidification/melting rates.

5.2. Changes of the shape of installed PCM plates on the internal temperature field of a refrigerated truck box

This study investigates how the shape of phase change cold storage materials affects the temperature field and system's cooling restoration capability inside the refrigerated truck box under varying door conditions. The study utilizes a refrigerated truck box equipped with PCM RT8HC, detailed in Section 3.1. However, in this study, the PCM shape is modified by implementing flow-guiding and distributing it to both sides of the box. Note that the total mass of PCM and the size of the refrigerated truck box remain unchanged. The mesh size, model assumptions, and solver settings in this section remain consistent with those in the original model.

5.2.1. Physical modeling of a medium-sized refrigerated truck box assembled with a conductive PCM

The overall framework of the physical model remains the same as the refrigerated truck box assembled with PCM as described in Section 3.1. However, modifications were made specifically to the shape of the PCM. The modified physical model is depicted in Fig. 19.

5.2.2. Analysis of numerical simulation results

Fig. 20 illustrates the top-view temperature distribution cloud diagram of the refrigerated truck box equipped with various structural PCMs, highlighting a 120-s monitoring period during the cooling phase after closure. The figure indicates that both conditions demonstrate a relatively rapid cooling rate in their respective flow-guiding areas of PCM. Flow-guiding PCM directs cold air to both sides of the box, resulting in a larger flow-guiding area compared to strip-shaped PCM. Moreover, the air temperature in the flow-guiding PCM condition is lower than that in the strip-shaped PCM condition on both sides of the box. This balanced temperature distribution ensures more uniform

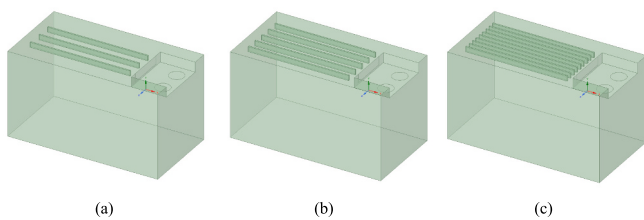


Fig. 14. Schematic shape of the box model of a medium-sized refrigerated truck assembled with different numbers of PCMs. (a) Assembly of 3 PCMs; (b) assembly of 5PCMs; (c) assembly of 9 PCMs.

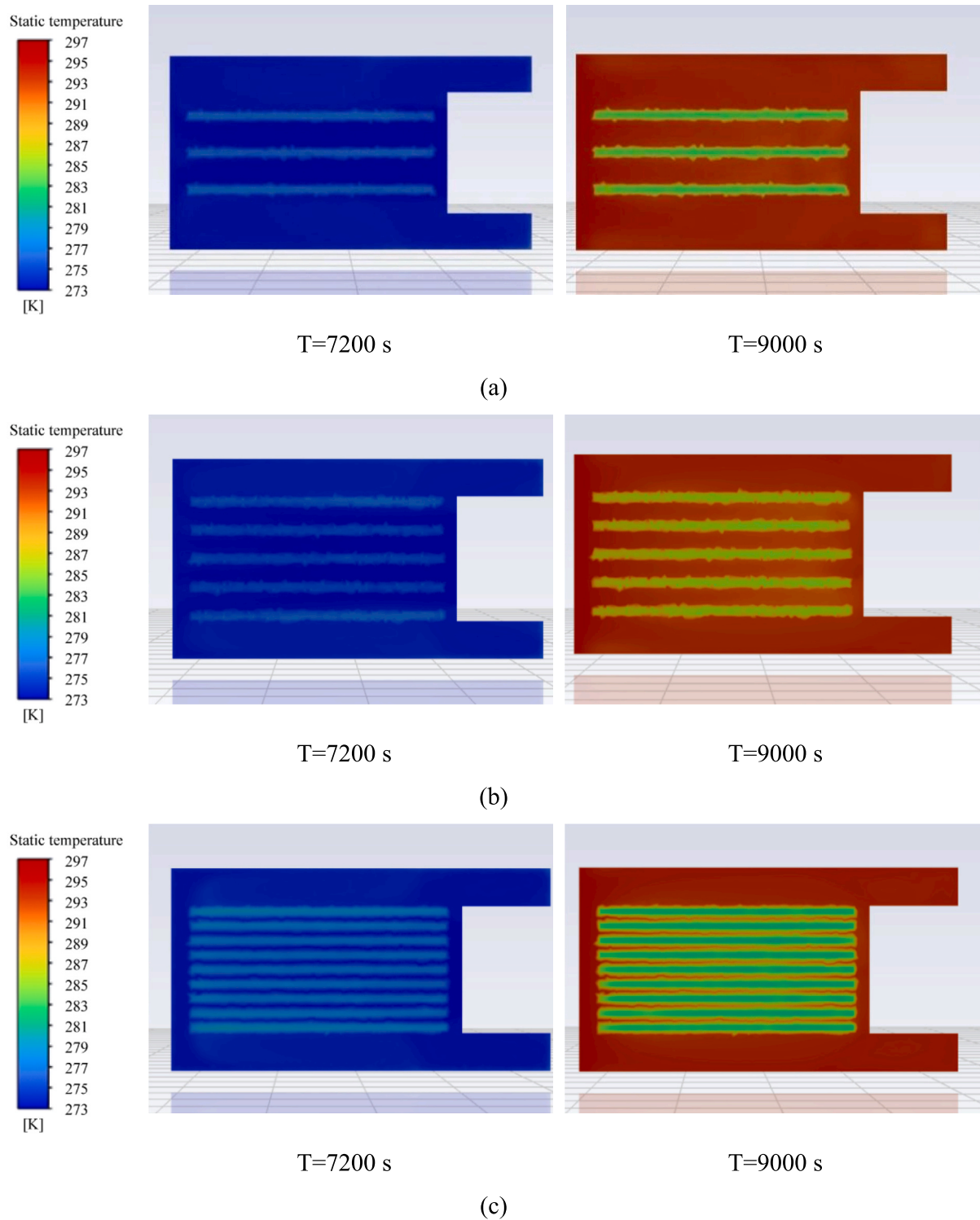


Fig. 15. Cloud view of temperature distribution of refrigerated trucks with different numbers of PCMs. (a) Assembly of 3 PCMs; (b) assembly of 5 PCMs; (c) assembly of 9 PCMs.

cooling in the refrigerated truck box, resolving the issue of slow cooling in certain non-windward areas.

Fig. 21 presents the liquid-phase rate change curves for two conditions: strip-shaped PCM and flow-guiding PCM. Both conditions initiate solidification simultaneously, but flow-guiding PCM demonstrates a higher solidification rate than strip-shaped PCM. At 7200 s, marking the conclusion of door closure transportation, the liquid-phase rate reaches its minimum value for both conditions. At this point, the solidification

efficiency of flow-guiding PCM is calculated to be 3.8 % higher than that of strip-shaped PCM. This improvement in heat transfer efficiency results from the reasonable modification of flow-guiding PCM shape while maintaining PCM mass unchanged. Strip-shaped PCM has a regular shape, resulting in relatively laminar airflow through its material center. In contrast, flow-guiding PCM alters airflow direction, creating more turbulent flow that enhances heat exchange between cold air and PCM, thereby improving material solidification rate.

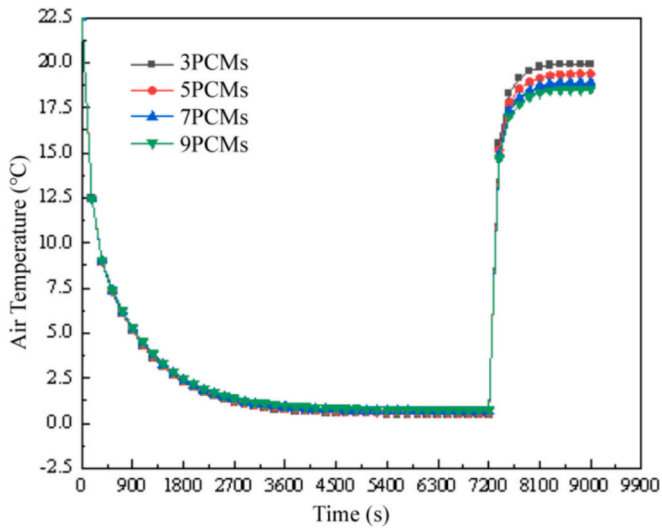


Fig. 16. Time-to-time variation curves of internal air temperature of refrigerated trucks fitted with different numbers of PCMs.

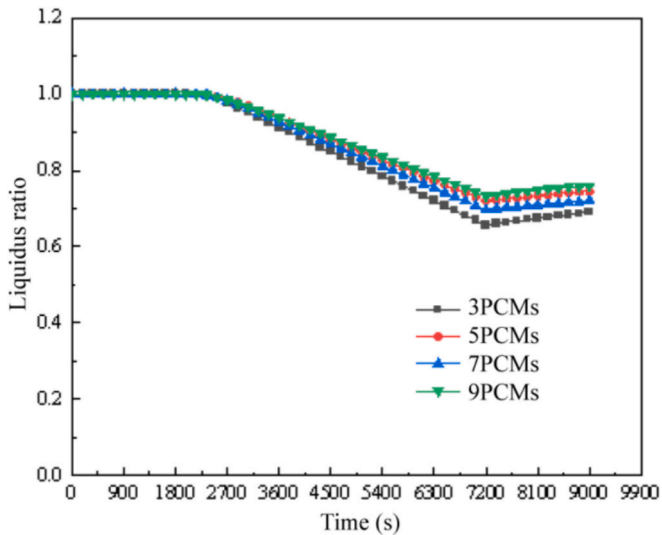


Fig. 17. Time-dependent variation curves of liquid phase rate for different quantities of PCMs.

6. Conclusions

In this paper, a comprehensive study of various aspects of the performance of a refrigerated truck box was carried out using Fluent. Firstly, the role of PCMs in medium-sized refrigerated trucks is affirmed by comparing two operating conditions containing PCMs with and without PCMs. Next, the effects of PCMs with different phase change temperatures were investigated while considering various factors such as PCM shape, number and installation location. In addition, the study delves into the effects of different numbers of PCMs on the temperature field inside the reefer box and the cooling capacity of the system during recovery, focussing on the operating conditions using the PCM RT8HC. Finally, the effects of two different PCM configurations, infusion and strip-shaped, on the temperature field inside the reefer box and the heat transfer efficiency of the PCM are investigated. The final conclusions of this paper are as follows:

- (1) During the first hour of the transportation cycle, the refrigeration system is active, and the air temperature inside the box stabilizes,

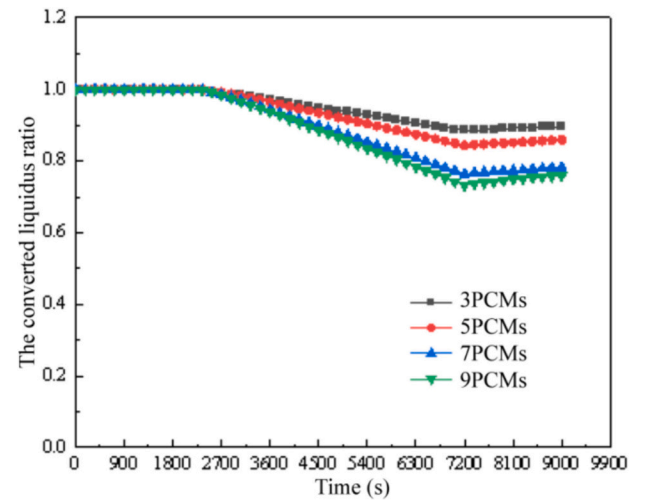


Fig. 18. Time-dependent variation curves of liquid phase rate after conversion of different amounts of PCMs.

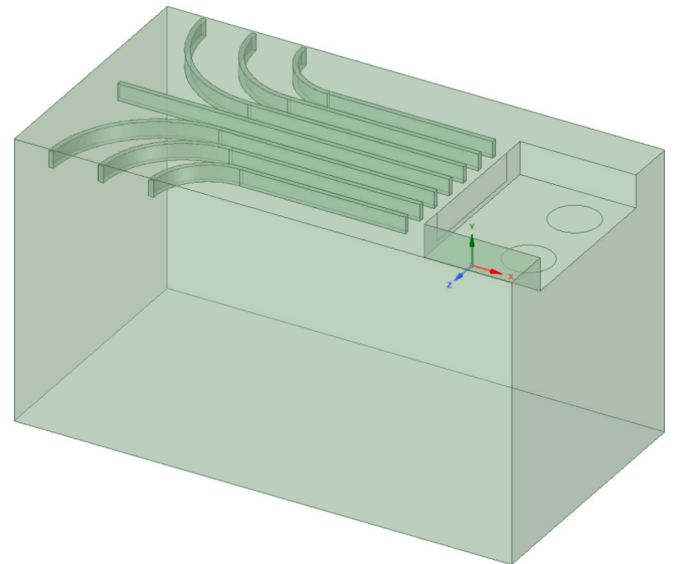


Fig. 19. Schematic shape of a medium-sized refrigerated truck box model assembled with a flow-guiding PCM.

resulting in a uniform temperature distribution. However, after two hours, when the cabinet door is opened, the temperature difference between the inside and outside of the box leads to the formation of natural convection. As a result, the higher temperature gas from the surroundings infiltrates into the box, causing the internal temperature to rise rapidly, the box is gradually presented as a “hot on the bottom of the cold” temperature field.

- (2) When the phase change cold storage material was added to the refrigerated truck box while keeping its original size unchanged, the system's ability to recover cooling was significantly improved, and the temperature fluctuation inside the reefer during the opening/closing period was reduced by 4 °C.
- (3) By comparing the results of these four working conditions, it was observed that the RT3HC working condition exhibited the greatest improvement in reducing the fluctuation of air temperature, making it the optimal performance among the four PCMs tested. Similarly, during the closing period, RT10HC showed the highest thermal efficiency, which was 87.56 % higher than RT3HC, but it stores excess cold in one transportation cycle.

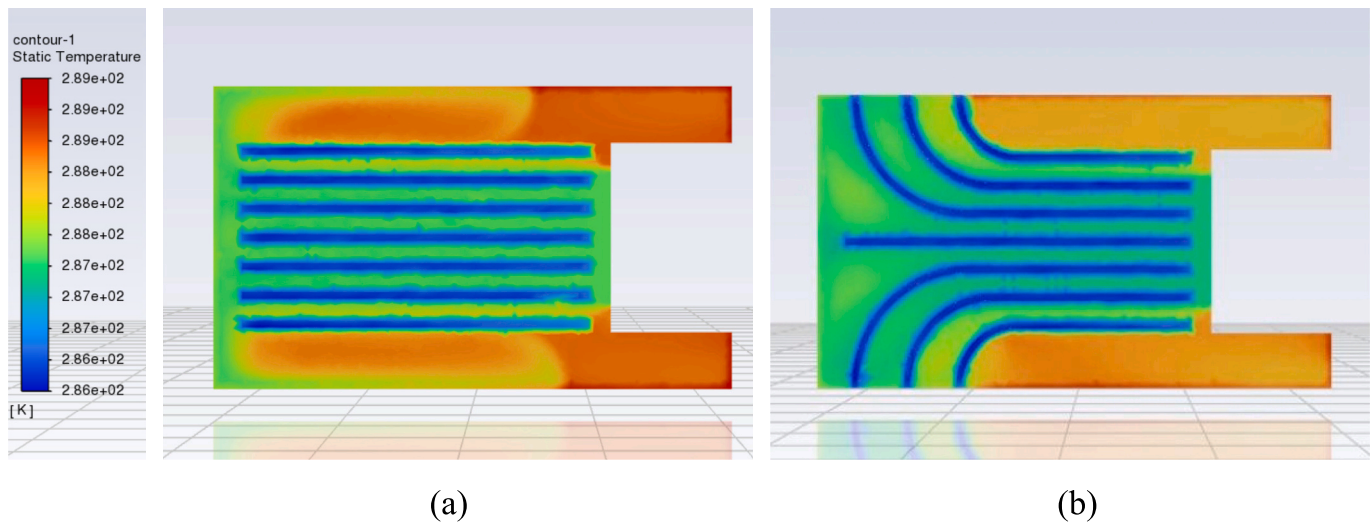


Fig. 20. Overhead temperature distribution of refrigerated truck box with different shapes of PCM. (a) Strip-shaped PCM (120 s); (b) flow-guiding PCM (120 s).

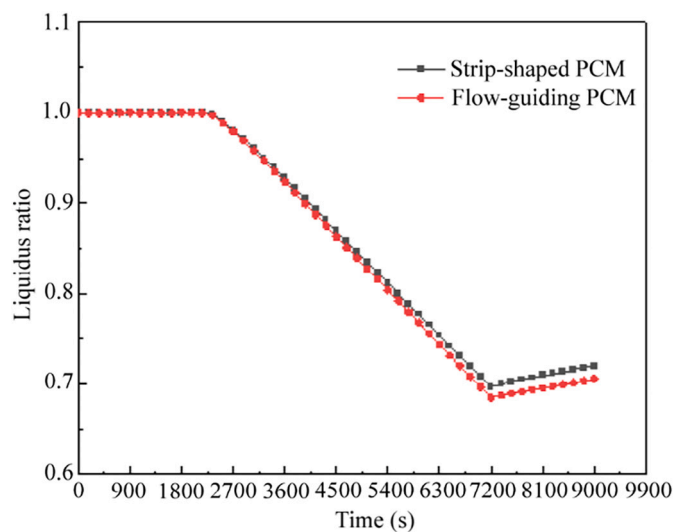


Fig. 21. Time-dependent variation curves of liquid phase rate for PCMs with different shapes.

- (4) The comparison of these four conditions showed that the 9 PCMs condition exhibited the greatest improvement in reducing air temperature fluctuation during both the door closing and door opening stages, making it the optimal choice compared to the 3 PCMs condition. By analyzing the effects of different quantities of PCMs, it was found that increasing the number of PCMs enhanced the heat transfer efficiency of the material and improved the solidification/melting rate. In particular, the heat transfer efficiency of the 9 PCMs condition was 57.13 % higher than that of the 3 PCMs condition.
- (5) The results showed that the flow-guiding PCM condition led to a more balanced temperature distribution inside the box. This solved the problem of slow cooling in some non-windward areas of the box. Additionally, the flow-guiding PCM condition exhibited improved heat transfer efficiency compared to the strip-shaped PCM condition. As a result, the solidification rate of the flow-guiding PCM was higher than that of the strip-shaped PCM, leading to a 3.8 % increase in solidification efficiency.

The subsequent efforts will concentrate on setting up an experimental platform for the medium-sized refrigerated truck box equipped

with PCM to validate the model by comparing experimental and simulation data. Additionally, numerical simulations and experimental investigations will be conducted on the cargo loading capacity parameter, aligning more closely with real-world refrigerated transport processes.

CRediT authorship contribution statement

Zhangyang Kang: Supervision, Formal analysis, Conceptualization. **Rufei Tan:** Writing – review & editing, Validation, Software. **Kaijie Qiu:** Writing – original draft, Software, Investigation, Data curation. **Qiongqiong Yao:** Visualization, Resources, Project administration. **Sen Liu:** Validation, Formal analysis, Data curation.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work. There is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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